UNIT - 2

SYNTHESIS AND FABRICATION

Bulk samples - Single crystals - Bottom-up - Top-down - Semiconductor nanostructures

1. Introduction - Two Approaches

Nanomaterials are made by two broad strategies:



Fig: Top-down vs bottom-up approach

	Top-down	Bottom-up
Principle	Bulk is reduced to nano size by mechanical / etching processes	Atoms / molecules / ions assemble into nano structures by chemical reactions
Examples	Ball milling, lithography, sputtering, etching, laser ablation	Sol-gel, hydrothermal, CVD, co-precipitation, self-assembly, MBE
Merits	Large-scale, good for patterns in electronics	Better size / shape control, fewer defects, cheaper, atomic precision
Demerits	Surface defects, contamination, wastage, costly equipment	Hard to scale and to position / interconnect nanoparticles

2. Synthesis of Bulk Polycrystalline Samples

A polycrystalline sample consists of many small randomly oriented crystallites (grains) separated by grain boundaries. It is the starting material for most ceramics, oxides, superconductors and thermoelectrics.

2.1 Solid-state reaction (ceramic method)

1. **Weighing**: high purity (99.9 %+) oxides / carbonates in stoichiometric ratio, e.g. $BaCO_3 + TiO_2$ for $BaTiO_3$.
2. **Mixing and grinding**: in agate mortar or ball mill with acetone / ethanol for 2 – 4 h so that the particles are intimately mixed (reaction occurs only at the contact points).
3. **Calcination**: heat at 0.6 – 0.8 T_m (e.g. 900 – 1100 °C) so that the carbonates decompose and the desired phase forms by inter-diffusion. $BaCO_3 + TiO_2 \rightarrow BaTiO_3 + CO_2$.
4. **Regrinding** and repeat calcination until a single phase is confirmed by XRD.
5. **Pelletising**: add 2 – 5 % PVA binder, press into discs in a die (~ 5 tonne) with a hydraulic press.
6. **Sintering**: heat pellet at 1200 – 1400 °C for hours; grains grow and pores close, densification up to 95 – 99 % of theoretical density.

> *Remember:*
> Merits: simple, cheap, bulk quantities. Demerits: high temperature, long time, inhomogeneity, large grains (micron size), volatile loss (Pb, Bi).

2.2 Other routes for polycrystals

- **Co-precipitation**: mix metal salt solutions and add precipitant ($NaOH$, NH_4OH) to precipitate hydroxides together; calcine to oxide. Gives atomic-level mixing and lower temperature
- **Sol-gel / Pechini / combustion synthesis**: citrate or glycine as fuel, self-propagating, produces fine powder in minutes
- **Arc melting / induction melting** for metals and alloys, intermetallics (under Ar atmosphere, remelting several times for homogeneity)
- **Hot pressing and Spark Plasma Sintering (SPS)**: pressure + pulsed current give dense nano-grained ceramics at lower T within minutes

3. Growth of Single Crystals

A single crystal has one continuous lattice without grain boundaries, needed for devices (Si wafers, lasers) and for studying anisotropic properties. Growth requires a supersaturated / supercooled state so nucleation and growth proceed in a controlled way.

3.1 Growth from melt

Used when the material melts congruently without decomposing.

(a) Bridgman - Stockbarger technique

- Material sealed in a pointed ampoule (quartz / graphite) is lowered slowly (1 - 10 mm/h) through a furnace temperature gradient from above T_m to below it
- Crystallisation starts at the tip (single nucleus, or seed) and proceeds upward
- Examples: GaAs, CdTe, CaF_2, LiF, alkali halides

(b) Czochralski (CZ) pulling method

- A seed crystal touches the melt in a crucible and is pulled upward (1 - 10 cm/h) while rotating (5 - 30 rpm); the crucible rotates in the opposite direction
- Melt temperature is held just above T_m; the crystal diameter is controlled by pull speed and temperature
- A neck (Dash necking) is first formed to eliminate dislocations; produces large, high-quality boules (up to 300 mm Si)
- Examples: Si, Ge, GaAs (LEC), sapphire, YAG, $LiNbO_3$

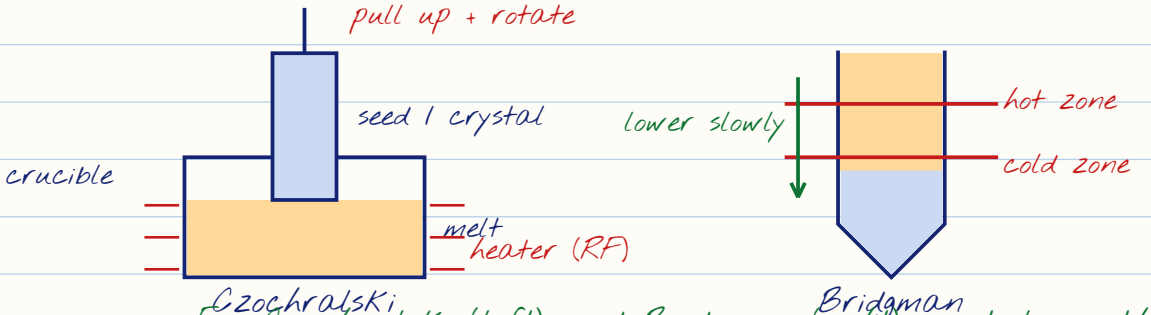


Fig: Czochralski (left) and Bridgman (right) crystal growth

(c) Other melt methods

- **Verneuil (flame fusion)**: powder falls through an oxy-hydrogen flame, melts and drops on a seed; synthetic ruby, sapphire, spinel

- **Zone melting / Float zone (FZ)**: a narrow molten zone travels along a rod; impurities segregate into the liquid (zone refining) so FZ-Si is ultra pure; no crucible contamination
- **Kyropoulos** method for large sapphire boules

3.2 Growth from solution

- **Low-temperature solution growth**: slow evaporation or slow cooling of a saturated solution (KDP, TGS, alum, NaCl); needs solubility with positive temperature coefficient, seed crystal in a constant temperature bath
- **Gel growth**: reactants diffuse slowly in silica gel; used for sparingly soluble compounds ($CaCO_3$, tartrates)
- **Flux growth**: material dissolved in a molten salt / oxide flux (PbO-PbF_2, Bi_2O_3) at high T and slowly cooled; for materials that melt incongruently (YIG, $BaTiO_3$)
- **Hydrothermal growth**: water at high T / P in an autoclave dissolves and re-crystallises material (quartz, ZnO) - see section 6

3.3 Growth from vapour

- **Physical vapour transport** (sealed tube sublimation), chemical vapour transport using a transport agent such as iodine: $MX + I_2 \rightleftharpoons MI_2 + X$, carried from hot to cold zone ($ZnSe$, TaS_2, MoS_2)
- **Epitaxial growth**: oriented overgrowth on a single crystal substrate (VPE, LPE, MBE, MOCVD)

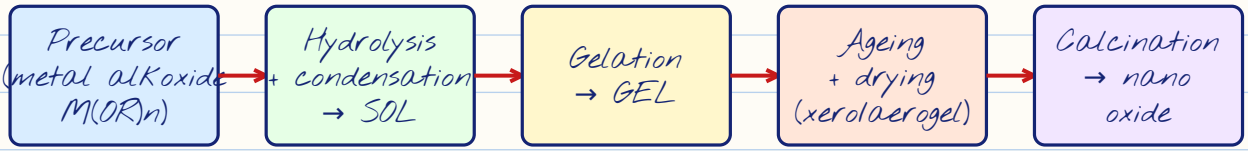
Method	Material suited	Advantages	Limitation
Czochralski	Si, Ge, GaAs, oxides	Large crystals, fast, high quality	Crucible contamination
Bridgman	Compounds, halides	Simple, volatile materials ok	Strain from the walls
Float zone	Si, refractory metals	Highest purity	Small diameter
Solution	Soluble salts, organics	Low T, low defect density	Slow (weeks)

Flux	Incongruent melting oxides	Growth below T_m	Flux inclusions
Vapour	Layered compounds, thin crystals	Pure, thin	Small size

4. Bottom-Up Approach

4.1 Sol-gel synthesis

A wet-chemical route in which a sol (colloidal particles dispersed in liquid) is converted into a gel (3D network containing liquid) and finally into a solid oxide.



H_2O, catalyst (acid / base), alcohol

Fig: Steps in sol-gel process

Reactions (e.g. TEOS / TTIP)

- **Hydrolysis**: $M\text{-}OR + H_2O \rightarrow M\text{-}OH + ROH$
- **Condensation (water)**: $M\text{-}OH + HO\text{-}M \rightarrow M\text{-}O\text{-}M + H_2O$
- **Condensation (alcohol)**: $M\text{-}OH + RO\text{-}M \rightarrow M\text{-}O\text{-}M + ROH$
- Final network of M-O-M bonds forms the gel. Example: $Si(OC_2H_5)_4 \rightarrow SiO_2$; $Ti(OC_3H_7)_4 \rightarrow TiO_2$
- **Effect of pH**: acid catalysis gives linear weakly branched chains (films, fibres); base catalysis gives highly branched dense particles (powders, Stober silica)
- Other factors: water-to-alkoxide ratio, temperature, solvent, precursor concentration
- **Drying**: ambient evaporation → xerogel (shrinks 5 - 10 times); supercritical drying → aerogel (very porous, density as low as 3 mg/cc, 99 % air, great insulator)
- **Coating techniques**: dip-coating, spin-coating (500 - 5000 rpm), spray; thickness ~ 50 - 500 nm per layer

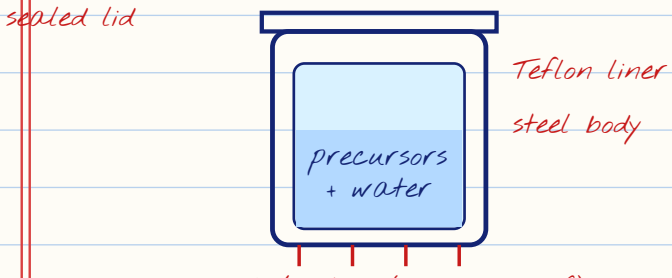
Advantages	Disadvantages

Low temperature, high purity and homogeneity, control of particle size / composition, films, fibres, monoliths, powders, cheap equipment	Expensive alkoxides, shrinkage and cracking on drying, long processing time, organic residues, hard to avoid hydroxyl groups

Applications: antireflective and self-cleaning coatings, catalysts, sensors, optical fibres, bioactive glass, aerogel insulation, ceramic membranes.

4.2 Hydrothermal (solvothermal) synthesis

Crystallisation of materials from aqueous solution at <r>high temperature (100 – 300 °C) and pressure (1 – 100 MPa)</r> in a sealed Teflon-lined stainless steel autoclave. If a non-aqueous solvent is used, it is called solvothermal.



1. Dissolve salts, adjust pH
2. Fill 60 – 80 % of liner
3. Heat 12 – 48 h; autogenous pressure
4. Cool, wash, dry the powder

Fig: Teflon-lined hydrothermal autoclave

- Under these conditions water has lower viscosity and dielectric constant, so solubility of oxides increases; the dissolved species nucleate and grow when supersaturated
- Size and shape (rods, wires, flowers, cubes) are tuned by temperature, time, pH, concentration and surfactants / capping agents (CTAB, PVP)
- Examples: ZnO nanorods, TiO_2 nanotubes, $BaTiO_3$, Fe_3O_4, quartz single crystals (growth at 400 °C, 1 – 2 Kbar), zeolites, MoS_2
- **Merits**: one-step, crystalline product without calcination, low T, uniform morphology, closed system (no volatile loss). **Demerits**: costly autoclaves, cannot observe the reaction, safety at high pressure, long time
- **Microwave-hydrothermal** heating reduces the reaction time to minutes

5. Thin Film Growth

A thin film is a layer of material with thickness from a few nm to about 1 μm deposited on a substrate. Properties depend on thickness, substrate, microstructure and deposition conditions.

Growth modes (nucleation and growth)

- **Frank-van der Merwe** (layer-by-layer, 2D): adatom-substrate bonding stronger than adatom-adatom, gives smooth epitaxial films
- **Volmer-Weber** (island, 3D): adatom-adatom bonding stronger, islands coalesce into a film, metals on insulators
- **Stranski-Krastanov**: layer-by-layer for a few monolayers, then islands due to strain (e.g. InAs on GaAs gives self-assembled quantum dots)

Steps: adsorption → surface diffusion → nucleation → island growth → coalescence → continuous film. Epitaxy = oriented growth, homo-epitaxy (Si on Si) or hetero-epitaxy (GaN on sapphire); lattice mismatch $f = (a_s - a_f)/a_f$ controls strain and dislocations.

Physical (PVD)	Chemical (CVD / solution)
Thermal / e-beam evaporation, sputtering, MBE, pulsed laser deposition (PLD), arc	CVD, MOCVD, PECVD, ALD, spray pyrolysis, spin / dip coating, electrodeposition, CBD, SILAR

5.1 Physical Vapour Deposition (PVD)

Material is converted into vapour by a physical means in high vacuum (10^{-5} - 10^{-7} torr), transported and condensed on the substrate without chemical reaction. Vacuum gives long mean free path and clean surface.

(a) Thermal / vacuum evaporation

- Source (in W / Mo boat or filament, resistive heating) heated until vapour pressure is high (~ 10^{-2} torr); atoms travel in a straight line (line of sight) to the substrate
- **E-beam evaporation**: focused 5 - 10 KeV electron beam heats a water-cooled crucible, for high melting materials (Ti, SiO_2, W)
- Deposition rate 1 - 10 nm/s, monitored by a quartz crystal

thickness monitor (frequency shift)

- Limits: alloy composition changes (different vapour pressures), poor step coverage

(b) Sputtering

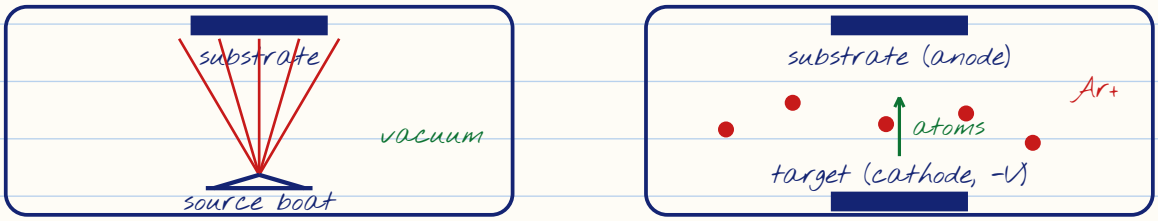
- Argon plasma (10^{-2} – 10^{-3} torr) is created between a target (cathode) and substrate (anode). Ar^+ ions are accelerated into the target and knock out atoms by momentum transfer; atoms condense on the substrate
- **DC sputtering** for metals; **RF sputtering** (13.56 MHz) for insulators / oxides; **Magnetron sputtering** uses magnets to trap electrons near the target, which raises ionisation and rate; **Reactive sputtering** with O_2 / N_2 to deposit oxides, nitrides (TiN, ZnO)
- Sputter yield depends on ion energy, mass and target; good adhesion and composition transfer; uniform large area films

(c) Pulsed Laser Deposition (PLD)

- Pulsed excimer laser (KrF 248 nm, 20 ns) is focused on a rotating target, creating an ablation plume that expands to the heated substrate in an oxygen background
- Stoichiometric transfer of complex oxides (YBCO, PZT, LSMO); good for epitaxial multi-layers; limitation: small area, droplets (particulates)

(d) Molecular Beam Epitaxy (MBE)

- Ultra high vacuum (10^{-10} torr). Effusion cells (Knudsen) produce atomic / molecular beams that reach the heated substrate without collisions; shutters switch the beams in < 1 s
- Growth rate ~ 1 monolayer/s (0.1 – 1 μm/h); **RHEED** shows oscillations giving in-situ monolayer control
- Used to make quantum wells, superlattices, quantum dots (GaAs / AlGaAs) and HEMT lasers. Very expensive and slow



Evaporation Sputtering

Fig: Schematic of evaporation and sputtering systems

5.2 Chemical Vapour Deposition (CVD)

A film is deposited by a chemical reaction of gaseous precursors on or near a heated substrate, volatile by-products are pumped away.

1. Precursor gases are transported into the reactor by a carrier gas (Ar / H_2).
2. They diffuse through the boundary layer to the hot substrate surface.
3. Adsorption → surface reaction / decomposition (e.g. $SiH_4 \rightarrow Si + 2H_2$ at 600 - 1000 °C).
4. Nucleation and film growth; by-products desorb and are exhausted.

Type	Key feature	Typical use
APCVD / LPCVD	Atmospheric / low pressure (0.1 - 10 torr) gives better uniformity	Poly-Si, SiO_2, Si_3N_4
PECVD	Plasma activates gas, so deposition at 200 - 400 °C	a-Si:H solar cell, passivation, DLC
MOCVD / MOVPE	Metal-organic precursors (TMGa, TMAl, NH_3)	GaN LEDs, laser diodes, solar cells
Thermal CVD	Hot wall / cold wall furnaces	CNT, graphene on Cu, diamond films
ALD	Self-limiting alternating pulses, one atomic layer per cycle	High-K Al_2O_3, HfO_2 gate oxide, conformal nanofilms

- **Advantages of CVD**: conformal coverage, high deposition rate, dense and pure, can coat 3D shapes, good for CNT / graphene / diamond growth

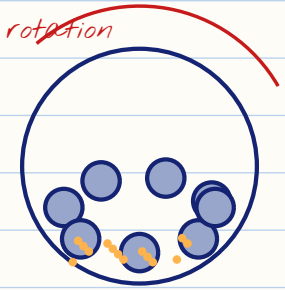
- **Disadvantages**: high temperature, toxic / flammable precursors, corrosive by-products
- **VLS growth** (Vapour-Liquid-Solid) of nanowires: Au catalyst droplet absorbs vapour (SiH_4) and supersaturates; the wire grows from the droplet interface. Wire diameter ≈ catalyst size

PVD	CVD
Physical process, no reaction	Chemical reaction at the surface
High vacuum, line of sight	Low to atmospheric pressure, conformal
Lower substrate temperature	Higher temperature (except PECVD)
Metals, alloys, simple compounds	Semiconductors, dielectrics, 2D materials

6. Top-Down Approach

6.1 Ball milling (mechanical attrition)

- A hardened steel / WC / zirconia vial is filled with balls (ball-to-powder weight ratio 10:1) and the powder; the vial rotates / vibrates at high speed (planetary mill: 200 – 600 rpm)
- Balls collide with the powder: repeated fracture, cold welding and re-fracture. After tens of hours grain size decreases to 5 – 20 nm
- Critical speed $N_c = 42.3/\sqrt{D}$ (rpm, D = mill diameter in m); operate at 65 – 75 % of N_c so that balls cascade (too slow no impact, too fast balls stick to the wall)
- **Mechanical alloying**: milling elemental powders to form alloys, amorphous phases, ODS superalloys, nanocomposites. Cryo-milling reduces recovery
- Process variables: milling time, speed, ball size / material, BPR, atmosphere (Ar), process control agent (stearic acid, methanol)



coarse powder → impact & shear
→ fracture / cold weld / re-fracture
→ nanocrystalline powder (5 - 20 nm)

Types: planetary, SPEX shaker, attritor

Fig: Ball milling

- **Merits**: simple, scalable, any material, low cost, dry process.
  Demerits: contamination from balls / vial, broad size distribution, agglomeration, strain and amorphisation

6.2 Micro fabrication

Set of techniques (from the IC industry) to create micron / nanometre patterned structures on a wafer: clean room (class 100 - 1000) → oxidation → deposition → lithography → etching → doping (diffusion / ion implantation) → metallisation → packaging.

- **Substrate preparation**: Si wafer cleaning (RCA clean: H_2O_2 + NH_4OH / HCl), thermal oxidation SiO_2 growth (dry / wet) as mask or insulator
- **Etching** removes unwanted material: **wet etching** (KOH, HF; isotropic or anisotropic along crystal planes) and **dry etching** (plasma RIE, DRIE Bosch process; anisotropic, vertical walls)
- **Doping** by ion implantation (energy 10 - 200 KeV, dose 10^{11} - 10^{16} cm^{-2}) + annealing
- **Bulk and surface micromachining**, LIGA, make MEMS / NEMS: cantilevers, accelerometers, micro-mirrors, microfluidic chips
- **Lift-off**: metal deposited over patterned resist, resist dissolves, leaving the metal only in openings

6.3 Lithography

Lithography (Greek: stone writing) transfers a pattern from a mask / design to a radiation-sensitive layer (resist) on a substrate.

1. **Substrate cleaning and priming** (HMDS for adhesion).
2. **Spin-coating the resist** (e.g. 1 μm at 3000 rpm), then soft bake (90 - 110 °C).
3. **Exposure** through a mask (or direct writing) to UV / e-beam /

X-ray.

4. **Development**: positive resist - exposed area dissolves; negative resist - exposed area cross-links and remains.
5. **Pattern transfer**: etch or deposit (lift-off); finally strip the resist.

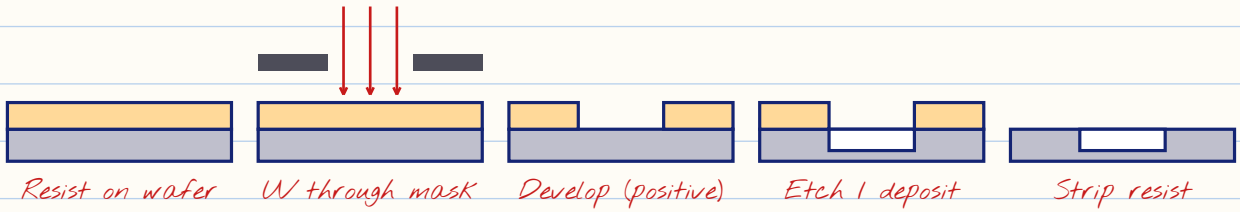


Fig: Photolithography steps

Type	Source	Resolution	Remarks
Photolithography	UV, DUV (248 nm KrF, 193 nm ArF), EUV 13.5 nm	down to ~ 5 - 7 nm node with EUV + multipatterning	Mass production, parallel, mask needed; diffraction limit $R = k_1 \lambda / NA$
Electron beam (EBL)	Focused electron beam, PMMA resist	~ 5 - 10 nm	Maskless, serial, slow, expensive; for masks and research
X-ray lithography	0.4 - 5 nm X-rays	< 50 nm	Large depth of focus, mask difficult, LIGA
Ion beam (FIB)	Ga^+ ions	~ 10 nm	Direct milling and deposition, TEM sample prep
Nano-imprint (NIL)	Mold pressed on resist, UV / thermal cure	< 10 nm	Cheap, high throughput, mold wear
Soft lithography / Dip-pen	PDMS stamp; AFM tip with ink	~ 30 - 100 nm	Printing of biomolecules, self-assembled monolayers

7. Requirements for Realising Semiconductor Nanostructures

To make quantum wells, wires and dots that really show quantum effects, the following requirements must be satisfied:

1. **Size / dimension control**: confinement length must be ≤ de Broglie wavelength or exciton Bohr radius (~ 5 – 20 nm), with size dispersion < 5 %, otherwise optical lines broaden (inhomogeneous broadening).
2. **High crystalline quality and interfaces**: atomically sharp, defect free interfaces; lattice matching (GaAs/AlGaAs $a \approx 5.65$ Å) or controlled strain; low density of dislocations and traps (which kill luminescence).
3. **Band gap engineering**: choose a pair of materials with suitable band offsets (ΔE_c, ΔE_v) so that carriers are confined in the narrow-gap material (type-I heterostructure).
4. **Purity and doping control**: ultrahigh vacuum, high-purity sources; **modulation doping** (dopants in the barrier) gives a 2D electron gas with very high mobility (> 10^7 cm^2/Vs).
5. **Surface passivation**: surface states must be removed by capping / shell (CdSe/ZnS core-shell, organic ligands) otherwise carriers recombine non-radiatively.
6. **Positioning, ordering and reproducibility** so devices can be integrated; low temperature measurement (carrier thermal energy $K_BT \ll$ level spacing).
7. **Stability** against oxidation and inter-diffusion at working temperature.

Structure	Common materials	Typical preparation
Quantum well / superlattice	GaAs/AlGaAs, InGaAs/InP, Si/SiGe	MBE, MOCVD (monolayer control)
Quantum wire	InP, Si, GaAs, ZnO nanowires	VLS CVD, MBE on V-grooves, e-beam lithography + etching
Quantum dot	CdSe, PbS, InAs, Si, InP, perovskite	Colloidal chemistry (hot injection), Stranski-Krastanov MBE / MOCVD, lithography

8. Growth Techniques for Nanostructures

8.1 Epitaxial growth methods

- **MBE** - UHV, atomic beams, highest control (monolayer). Quantum wells, superlattices, self-assembled QDs
- **MOCVD / MOVPE** - metal-organics + hydrides at 500 - 800 °C, fast and scalable, GaN LEDs, laser diodes, fewer cost per wafer
- **Liquid phase epitaxy (LPE)** - film grows from a supersaturated melt on a substrate, thick films, low cost, simple
- **Chemical beam epitaxy (CBE)** - metal organic beams in UHV (hybrid)
- **Atomic layer epitaxy / ALD** - one atomic layer per cycle

8.2 Self-assembled quantum dots (Stranski-Krastanov)

When a lattice-mismatched film (InAs on GaAs, mismatch 7 %) grows beyond a critical thickness (~ 1.7 monolayer) strain relaxes by forming coherent dislocation-free islands on a thin wetting layer. Dots of 10 - 30 nm base, 3 - 10 nm height with density 10^{10} - 10^{11} cm^{-2} are obtained and can be capped by GaAs.

8.3 Vapour-Liquid-Solid (VLS) for nanowires

Metal catalyst (Au) forms a liquid eutectic alloy droplet with the semiconductor (Au-Si eutectic at 363 °C). Precursor vapour dissolves in the droplet, on supersaturation, solid precipitates under the droplet, extending a wire. Diameter is controlled by the droplet size, also vapour-solid (VS) and solution-liquid-solid (SLS) variants.

8.4 Colloidal synthesis of quantum dots

- **Hot injection** (Murray, Bawendi): a cold solution of precursor (e.g. TOP-Se) is injected into a hot (250 - 300 °C) solution of Cd precursor with coordinating ligands (TOPO, oleic acid). A burst of nucleation is followed by slow growth at a lower temperature (LaMer model)
- Size controlled by time and temperature, ligands (capping agents) stop aggregation
- **Core-shell** (CdSe/ZnS) to improve photoluminescence yield and

stability

- **Ostwald ripening**: larger particles grow at the cost of smaller ones; narrow size distribution requires nucleation and growth to be separated

> Summary:
> Top-down gives patterned structures of 10s of nm but with edge damage; bottom-up epitaxy and colloidal chemistry give atomically precise, defect-free structures. Most devices combine both.

Quick Revision - Important Questions

1. Describe solid-state reaction method for the synthesis of polycrystalline sample.
2. Explain Czochralski and Bridgman methods of single crystal growth with neat diagrams.
3. Explain the sol-gel process: steps, chemistry, advantages and applications.
4. Describe hydrothermal synthesis with a diagram.
5. Distinguish PVD and CVD. Explain sputtering / evaporation and MOCVD.
6. Explain the principle of ball milling and its merits / demerits.
7. Describe photolithography and e-beam lithography.
8. What are the requirements for realising semiconductor nanostructures? Discuss MBE and VLS growth.